Salt Crystal Impurity Detection and Quality Grading Using Computer Vision for Tropical Salt Production

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***Abstract*—Salt production in tropical regions, such as Sri Lanka, relies predominantly on manual visual inspection for quality assurance. This approach is inherently subjective, inconsistent, and throughput constrained. The present study describes the design, implementation, and evaluation of a real-time computer vision system that automates impurity detection and quality grading in salt crystals through deep learning techniques. The system uses a YOLOv8 object detection model trained on a custom-annotated dataset of 1,952 salt crystal images, categorized as pure salt, impure salt, or unwanted foreign material. The trained model is exported in ONNX format and integrated into a distributed microservices architecture, which includes a NestJS backend with gRPC-based inference, a MongoDB persistence layer, and a Next.js frontend dashboard for real-time visualization and compliance reporting. A quantitative grading algorithm calculates per crystal whiteness indices using HSV color space analysis and generates composite quality scores by combining detection confidence with spectral purity metrics. Experimental evaluation on a held-out test set demonstrates a precision of 99.41%, a recall of 99.14%, a mean Average Precision (mAP@0.5) of 99.37%, and a mAP@0.5:0.95 of 79.95%. Per-class analysis verifies robust detection performance across all three categories. Confusion matrix analysis shows particularly strong results for impure salt detection (98.86% precision, 99.67% recall), the most operationally critical class. This study is among the first to report on a dedicated AI-based quality control system developed for tropical salt production environments.**

Keywords—computer vision, deep learning, YOLOv8, salt quality inspection, object detection, convolutional neural networks, industrial quality control, real-time image processing. Introduction

I. INTRODUCTION

Salt is among the most widely used mineral commodities, serving essential roles in food preservation, chemical production, pharmaceutical manufacturing, and water treatment [1]. Global production exceeds 300 million metric tons annually, with tropical and subtropical countries relying primarily on solar evaporation [2]. In Sri Lanka, the Puttalam coastal region produces over 150,000 metric tons per year through traditional open-air salterns whose crystallization cycles are tightly coupled with seasonal monsoon patterns [3]. The economic significance of this industry extends beyond domestic consumption, as Sri Lankan salt enters regional export markets where compliance with international quality standards is essential for market access.

Despite this scale, quality assurance methods in tropical salt production remain predominantly manual. Trained operators visually inspect salt crystals under ambient or semi-controlled lighting to identify contaminants including clay particles, sand grains, dark mineral inclusions, organic debris, and microbial discoloration [4]. This dependence on human visual assessment introduces several well-documented limitations: inter-operator variability leading to inconsistent grading outcomes, fatigue-induced degradation of detection sensitivity during extended shifts [5], throughput constraints imposed by human cognitive processing speed, and the inability to generate quantitative data necessary for statistical process control and regulatory compliance documentation.

Recent advances in deep learning and high-resolution imaging have enabled automated visual inspection across diverse industrial domains, including surface defect detection in manufacturing [6], agricultural quality grading [7], and semiconductor wafer classification [8]. The YOLO family of single-stage object detectors has achieved a favorable balance between accuracy and speed for real-time deployment [9]. YOLOv8, the latest iteration, introduces an anchor-free detection head, decoupled prediction branches, and an enhanced feature pyramid network that improve performance on irregularly shaped and multiscale objects [10]. However, no existing system applies deep learning-based object detection to salt crystal quality control in tropical production environments, where challenges include variable outdoor illumination, crystal translucency, specular reflectivity, sub-millimeter contaminant dimensions, and subtle visual differences between acceptable crystalline variations and genuine defects.

This paper addresses this gap by presenting the design, implementation, and evaluation of a real-time computer vision system integrating YOLOv8-based detection with quantitative quality grading within a production-grade distributed architecture. The principal contributions are: (1) the first AI-based quality inspection system for tropical salt production; (2) real-time three-class impurity detection using YOLOv8 trained on 1,952 custom-annotated images; (3) a multi-parameter grading algorithm combining HSV-based whiteness indices with classification-weighted composite quality scores; and (4) integration with a cloud-accessible

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monitoring dashboard providing real-time visualization, batch management, and compliance reporting.

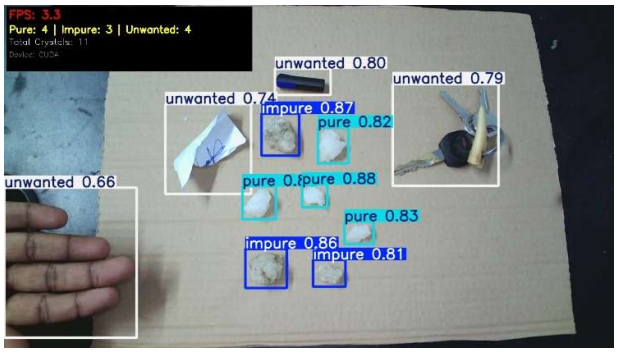


Figure I: Validated Output from YOLOv8

II. Literature Review

The application of computer vision and deep learning to automated quality inspection has rapidly expanded across various industrial sectors over the past decade. However, adoption levels vary significantly across industries, and notable gaps remain in mineral product quality analysis. This section critically reviews existing research across four key themes to position the present work within the current state of knowledge.

A. Deep Learning in Industrial Quality Inspection

The transition from handcrafted feature engineering to end-to-end deep learning pipelines has fundamentally transformed industrial quality control. Ren. [6] demonstrated that CNN-based surface inspection frameworks can generalize effectively across multiple industrial datasets through transfer learning, even with limited labeled training data. He. [14] advanced this direction by introducing multiscale hierarchical feature fusion for steel surface defect detection, establishing that combining features extracted at different network depths enables simultaneous detection of both fine-grained and large-scale anomalies a principle directly applicable to salt crystal inspection where impurity sizes range from microscopic discoloration to visible foreign bodies. Luo. [15] synthesized findings from over 120 publications on steel surface defect detection and confirmed that CNN architectures consistently outperform traditional methods in both accuracy and robustness across varying imaging conditions. These collective findings establish that learned visual representations surpass manually designed features for industrial inspection tasks, yet nearly all existing work targets manufactured products with controlled defect taxonomies rather than natural crystalline materials produced in uncontrolled outdoor environments.

B. Computer Vision in Food and Agricultural Quality Evaluation

Food and agricultural quality grading systems offer methodological parallels to salt crystal inspection. Li. [7] employed CNNs for apple quality classification and achieved 95.33% accuracy, substantially outperforming conventional HOG and GLCM-based approaches and validating the superiority of learned feature representations for natural product grading. Bhupendra. [16] developed a deep CNN system for classifying milled rice grains into seven damage categories using high-magnification imagery, attaining 98.37% accuracy with EfficientNetB0. The parallels between grain classification and salt quality assessment are notable: both involve small object dimensions, subtle inter-class visual differences, and sensitivity to illumination conditions. Mukhiddinov. [17] further extended YOLO-based architectures to classify fruit and vegetable freshness, bridging the gap between object detection and food quality grading. However, none of these agricultural methods address crystalline or translucent materials, which introduce unique optical complexities including specular reflections, internal light scattering, and partial transparency that fundamentally alter the visual appearance of defects.

C. YOLOBased Object Detection for Quality Control

The YOLO family of single-stage detectors has become a preferred architecture for real-time industrial inspection owing to its favorable speed-accuracy tradeoff. Terven. [10] comprehensively reviewed YOLO architectures from YOLOv1 through YOLOv8 and YOLO-NAS, identifying anchor-free detection heads, decoupled prediction branches, and improved feature pyramid networks as the key innovations enabling detection of irregular shapes across multiple scales. Xiao. [18] applied YOLOv8 to fruit ripeness identification and achieved 99.5% accuracy, demonstrating the effectiveness of the C2f module and anchor-free design for fine-grained classification tasks. Within the mineral domain, Rochman. [13] classified salt quality from Indonesian tropical salterns into four grades using machine learning, achieving 91.7% accuracy with K-Nearest Neighbours. However, their approach relies on chemical analysis parameters rather than visual inspection, fundamentally limiting its applicability for real-time production line deployment. Li. [12] demonstrated the viability of attention-enhanced CNNs for mineral image classification, confirming that deep learning can distinguish visually similar crystalline materials based on texture, color, and surface morphology features.

Critically, no existing study combines real-time object detection with spatial localization of impurities within crystal images, nor integrates such detection with quantitative grading metrics including whiteness indices and composite quality scores. Furthermore, no published system addresses the specific environmental conditions of tropical outdoor salt production variable solar illumination, high humidity, and the complex optical behavior of translucent salt crystals. This unaddressed gap directly motivates the architecture and methodology presented in this paper.

III. Methodology

This section outlines the end-to-end methodology used for developing the proposed quality control system. The approach includes preparing datasets, selecting and training model architecture, defining quality metrics, and establishing the deployment pipeline.

A. Dataset Collection and Annotation

A custom image dataset was constructed for this research, as no publicly available annotated dataset of salt crystal images exists. Raw images were captured at multiple solar evaporation saltern sites in the Puttalam coastal region of Sri Lanka using high-resolution digital cameras positioned at fixed distances above salt crystallization beds and sample trays. Imaging sessions were conducted across different times of day to capture natural variations in ambient lighting intensity, direction, and color temperature, including direct

midday sunlight, overcast diffuse illumination, low-angle morning and evening lighting, and supplementary LED illumination.

Each image was manually annotated using the LabelStudio and CVAT annotation tools. Bounding box annotations identified individual salt crystals and contaminant particles, with each region assigned to one of three target classes: pure salt, which represents clean crystals with good whiteness and clarity; impure salt, indicating crystals with visible discoloration, mineral inclusions, or surface contamination; and unwanted material, covering foreign objects such as sand grains, clay particles, organic debris, and non-salt mineral fragments. The initial manually annotated dataset included approximately 650 images. The annotation quality was verified through a secondary review to ensure spatial accuracy and correct class assignment.

Table 1: Dataset Configuration for Training

Parameter	Values
Total images (Original + Augmentted)	1952
Training Set	1366 (70%)
Validation Set	293 (15%)
Test Set	293 (15%)

B. Data Augmentation Strategy

A custom dataset was constructed since no publicly available annotated salt crystal dataset exists. Raw images were captured at multiple solar evaporation saltern sites in the Puttalam coastal region of Sri Lanka under diverse lighting conditions including direct midday sunlight, overcast diffuse illumination, low-angle morning and evening lighting, and supplementary LED illumination. Each image was manually annotated using LabelStudio and CVAT with bounding boxes assigned to three target classes: pure salt (clean crystals with good whiteness and clarity), impure salt (crystals with visible discoloration, mineral inclusions, or surface contamination), and unwanted material (foreign objects including sand, clay, organic debris, and non-salt fragments). Annotation quality was verified through secondary review for spatial accuracy and correct class assignment.

The initial dataset of approximately 650 images was expanded through systematic augmentation using the Albumentations library [17] and native YOLOv8 tools. Applied transformations included random rotation (±15°), horizontal and vertical flipping, brightness adjustment (±20%), contrast modification, Gaussian noise injection, mosaic augmentation creating composite images from four source samples, and cutout augmentation to encourage distributed feature learning. The final dataset comprised 1,952 annotated images partitioned into training (70%, 1,366 images), validation (15%, 293 images), and test (15%, 293 images) subsets using stratified sampling.

C. YOLOv8 Model Architecture

YOLOv8 was selected for its capacity to perform spatial localization and categorical classification in a single forward pass, a requirement for computing per-crystal quality metrics and generating visual overlays for operator monitoring [10]. The architecture comprises a CSPDarknet53 backbone with Cross Stage Partial connections and the C2f (cross-stage partial bottleneck with two convolutions) module for multi-scale feature extraction, a Spatial Pyramid Pooling Fast (SPPF) layer for fixed-size multi-scale feature aggregation, an improved Path Aggregation Network (PANet) neck enabling bidirectional feature fusion across pyramid resolutions, and a decoupled anchor-free detection head that processes objectness, classification, and bounding box regression through separate branches. The anchor-free design is particularly advantageous for salt crystal detection because both crystals and impurities exhibit highly irregular shapes that do not conform to the fixed aspect-ratio assumption. The YOLOv8m (Medium) variant with approximately 3.2 million parameters was selected to balance detection accuracy with the real-time processing requirement of five frames per second.

Training was conducted using the Ultralytics framework with Python 3.10 and PyTorch. The model was initialized with COCO-pretrained weights for transfer learning, fine-tuned for 150 epochs with an initial learning rate of 0.01, cosine-annealed scheduling, and batch size 16 at 320×320 input resolution. The composite loss function integrates CIoU bounding box regression loss [18], objectness loss, and classification cross-entropy loss. Early stopping monitored validation mAP@0.5 with a patience of 20 epochs. The best checkpoint was exported to ONNX format with fixed input shape 1×3×320×320 and operator set version 13 for runtime-independent deployment through ONNX Runtime.

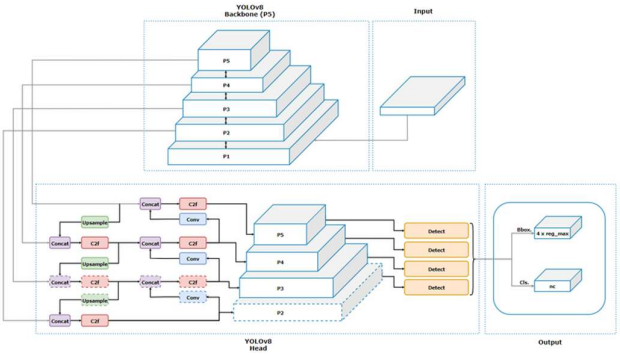


Figure II: YOLOv8 Model Architecture

D. Quality Metrics Formulation

Beyond categorical detection, the system implements a multiparameter quality assessment framework consisting of three quantitative metrics.

Purity Percentage. The batch purity is calculated as the ratio of pure crystal detections to total salt crystal detections, expressed as:

$$\boldsymbol{Purity} = \frac{\boldsymbol{N_{Pure}}}{\boldsymbol{N_{Pure} + N_{Impure}}} \times \mathbf{100} \qquad (01)$$

where N_{pure} and N_{impure} denote the count of pure and impure salt detections, respectively. Detections classified as unwanted material are excluded from the denominator as they represent nonsalt foreign objects.

Whiteness Index. For each detected bounding box region, the system extracts the pixel content, converts it to HSV color space representation, and computes a per-pixel whiteness value as the product of the Value channel and the

complement of the Saturation channel. The region whiteness score is the mean of these per-pixel values:

$$\boldsymbol{W} = \left(\frac{\mathbf{1}}{\boldsymbol{n}}\right) \times \sum(\boldsymbol{V_i} \times (\mathbf{1} - \boldsymbol{S_i})) \times \mathbf{100} \quad (02)$$

where $\boldsymbol{V_i}$ and $\boldsymbol{S_i}$ are the normalized Value and Saturation of pixel *i*, and *n* is the total pixel count within the bounding box. This formulation captures the dual requirement that high-quality salt crystals exhibit both high brightness (high V) and low color saturation (low S).

Composite Quality Score. A unified quality indicator for each detection is computed as:

$$\boldsymbol{Q} = \boldsymbol{W} \times \boldsymbol{a_c} \quad (03)$$

where *W* is the whiteness index and $\boldsymbol{a_c}$ is a class-dependent weighting factor: 1.0 for pure salt, 0.3 for impure salt, and 0.0 for unwanted material. This composite metric integrates spectral quality with classification outcome into a single quantitative indicator suitable for automated threshold-based batch acceptance or rejection decisions.

IV. Results And Discussion

This section shows the experimental results from testing the proposed system on the held-out test partition, followed by a comparison with related work and a discussion of the findings and limitations.

A. Evaluation Metrics

Model performance was assessed using standard object detection metrics: Precision, Recall, mAP@0.5, and the stricter mAP@0.5:0.95, which averages AP across IoU thresholds from 0.5 to 0.95 in steps of 0.05. All metrics were computed using the Ultralytics evaluation pipeline on the 15% held-out test set, which was not exposed during training or validation.

B. Overall Detection Performance

The trained YOLOv8n model achieved a precision of 99.41%, recall of 99.14%, mAP@0.5 of 99.37%, and mAP@0.5:0.95 of 79.95%. These results confirm that the model detects and classifies salt crystals and contaminants with near-perfect reliability at the standard IoU threshold. The mAP@0.5:0.95 score of 79.95% reflects the expected decline under stricter localization requirements, consistent with the irregularly shaped boundaries of natural salt crystals that make pixel-precise bounding box alignment inherently challenging.

Table 2: Overall Model Detection Performance

Metric	Value	Description
mAP@50	0.9937	Mean Average Precision @ IoU 50%
mAP@50-95	0.7995	Mean AP across IoU thresholds
Precision	0.9941	Correct detections ratio
Recall	0.9914	Detected objects ratio

C. Per-Class Performance Analysis

Analysis of the confusion matrix across 1,588 detections from 293 test images reveals distinct characteristics for each class. The pure salt class achieved 96.32% precision and an ideal recall of 100%, meaning all pure crystals were correctly identified. The 25 false positives mainly arose from background areas and light-colored foreign particles that resembled clean crystals. The impure salt class had the highest precision at 98.86% and a recall of 99.67%, with only 2 out of 609 impure crystals missed. This is especially important because detecting impure salt is the most operationally critical capability, as undetected impurities can directly affect product quality. The unwanted material class showed the lowest precision at 92.47% and recall at 96.43%, with 21 of 22 false positives coming from background regions misidentified as contaminants. This class is the most visually varied, including sand grains, clay particles, and organic debris, which explains the wider range of errors.

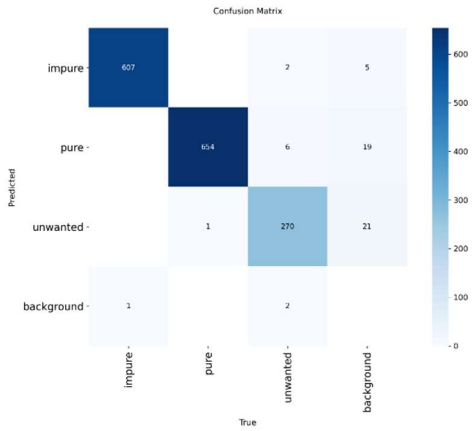


Figure III: Model Confusion Metrix

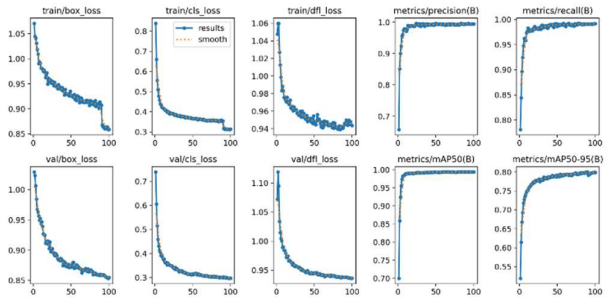


Figure IV: Model Training Results

D. Limitations

Several limitations should be recognized. The dataset was gathered from a single geographic area (Puttalam, Sri Lanka); the ability to apply findings to salt from other regions with different mineral compositions and contaminant profiles has not been tested. The system analyzes frames independently without tracking objects over time adding algorithms like DeepSORT could improve accuracy by combining observations across frames. The confusion matrix shows that background false positives mainly affect all categories, indicating that better region of interest setup and background suppression could improve precision. Also, the system now

acts as a monitoring tool without physical actuation for automatic batch rejection, which would be needed for fully autonomous closed-loop quality control.

E. Comparison with Related Work

The proposed system substantially outperforms Rochman's [15] 91.7% salt quality classification accuracy, which relied on chemical features rather than visual detection. Xiao's [14] 99.5% fruit ripeness detection with YOLOv8 confirms the architecture's suitability, with results broadly consistent with this study. Uniquely, the present system integrates spatial detection with quantitative per-crystal quality metrics (whiteness index, composite quality score) and a production-deployable real-time monitoring infrastructure combination, no prior work has offered for mineral product quality control.

V. Conclusion And Future Work

This study presented the design, implementation, and evaluation of a real-time AI-powered computer vision system for automated detection of salt crystal impurities and quantitative quality grading in tropical production environments. The system utilizes a YOLOv8 object detection model trained on a custom dataset of 1,952 annotated salt crystal images to detect and classify crystals into three operational categories: pure salt, impure salt, and foreign material. The model is integrated within a distributed architecture consisting of a NestJS backend with ONNX Runtime inference via gRPC, a MongoDB database for inspection data storage, and a Next.js dashboard that enables real-time detection visualization, batch management, analytics, and exportable quality reports.

Experimental results demonstrate strong detection performance on a held-out test set, achieving 99.41% precision, 99.14% recall, 99.37% mAP@0.5, and 79.95% mAP@0.5:0.95 while maintaining real-time processing at five frames per second with sub-200 ms inference latency. Class-level evaluation further showed that the critical impure salt category achieved 98.86% precision and 99.67% recall, indicating highly reliable contamination detection suitable for practical deployment in industrial production environments.

In addition to detection, the proposed system introduces a quantitative quality grading framework combining purity percentage, HSV-based whiteness index, and a class-weighted composite quality score. These metrics enable automated batch acceptance decisions, statistical process monitoring, and regulatory compliance reporting, replacing subjective manual inspection with consistent and data-driven evaluation. Overall, the proposed approach outperforms previous machine-learning-based salt grading methods and demonstrates the potential of computer vision systems to modernize mineral product quality control through accurate, scalable, and fully automated inspection.

Future work includes integration with conveyor belt systems and automated rejection actuators for closed-loop quality control, edge computing deployment on NVIDIA Jetson platforms for remote saltern locations with limited connectivity, dataset expansion with geographically diverse salt samples to improve model generalizability, incorporation of temporal object tracking algorithms for per-crystal observation aggregation across consecutive frames, and adaptation of the system architecture for quality inspection in adjacent granular material industries including sugar processing, rice milling, and fertilizer production

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